

CHEIKH TOURE

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EDUCATION

NORTH CAROLINA AGRICULTURAL & TECHNICAL STATE UNIVERSITY

Greensboro, NC

Electrical Engineering

Aug 2025 – May 2029

GPA – 3.84

Relevant Coursework: Electric Circuit Analysis, Instrumentation & Networks Lab, Differential Equations, Calculus III

SKILLS

MATLAB, Arduino, Python, Embedded Systems, Microcontroller Programming, Sensor Integration, Real-Time Monitoring & Control, Circuit Analysis

PROJECTS

MEDREACH

NCAT HackNat 9.0

Team of 4

Nov 2025

- Secured 2nd Place Award for innovation out of 10+ competing teams by engineering a Python-based teleoperation system for a 6-DOF surgical robot simulation.
- Extended the concept by duplicating the rod to mimic the fingers of a human hand, allowing full-range human-like control in real time.
- Enabled full-motion playback analysis for training review by architecting a JSON telemetry system that logged and reconstructed joint data sequences.

FLOW FREE ML SOLVER – UNDERGRADUATE RESEARCH

NC A&T ADEPT Lab

Python · OpenCV · NumPy · TensorFlow

Feb 2026 – Apr 2026

- Collaborated with a 7-member research team advised by Dr. Daniel Limbrick to explore machine learning solutions to a constraint-based puzzle with direct ties to VLSI wire routing in circuit design.
- Built an image processing pipeline using OpenCV to convert Flow Free screenshots into structured integer matrices via contour detection and HSV color thresholding.
- Presented a co-authored research poster at the NC A&T Undergraduate Research Symposium.

SMART PARKING ASSISTANT

Apr 2026

- Designed and programmed a real-time distance-sensing control system using an HC-SR04 ultrasonic sensor and an ELEGOO UNO R3 microcontroller, delivering automated visual and auditory feedback based on continuous proximity zone monitoring

EXPERIENCE

WELL SPRING RETIREMENT COMMUNITY

Greensboro, NC

Diet Tech

Jun 2024 - Present

- Maintained 90%+ compliance with regulatory health and safety standards during yearly inspections through systematic inventory tracking, protocol adherence, and standards verification.
- Communicated status updates and changes clearly to staff and care teams, ensuring accurate documentation and continuity of care